Is the platform still there?

**Monitor presence at the system level,
not separately for every message stream.**

**Discovery keeps track of participants.
RouterHealth tells the application what we are hearing.**

Three choices, three different jobs

Choice	Plain-language meaning	Fit for our goal
Reader / writer liveliness	"This writer is still there."	Too narrow for whole-platform presence; finite writer leases disable writer-side filtering in Connext 7.7
Participant liveliness / SPDP2	"Keep this participant in the discovery database."	Maintains discovery records and removes expired participants
Periodic RouterHealth + deadline	"We are hearing from this platform, with fresh metadata."	System-presence signal with application callbacks

Use each for its own job. Do not make every data writer a platform heartbeat.

SPDP2 keeps the discovery database current

Discovery is the middleware's list of known participants.

- SPDP2 sends periodic assertions: "This participant is still here."
- Receiving assertions keeps the participant's discovery record alive.
- If its discovery lease expires, the middleware removes the participant.
- This supports discovery and cleanup; it is not our application-health report.

> **SPDP2 answers:** "Should this participant remain in discovery?"
>
> **Our application asks:** "Is the platform still reachable, and what is its state?"

RouterHealth tells the application what changed

```
Platform sends periodic RouterHealth metadata
                  |
                  v
Receiver gets an update, or a deadline callback reports a miss
                  |
                  v
Presence policy updates the platform state and notifies the application
```

- **Updates arrive:** refresh platform identity, health summary, and presence.
- **Updates stop:** mark stale; mark down after the agreed grace period.
- **Updates resume:** report recovery using current platform / session evidence.

> **"Down" means unreachable from this observer, not proven physically dead.**

Validate all wire traffic by role and purpose

For every traffic type: who sends it, who needs it, and why is it on the wire?

Traffic	Its job	What to validate
Discovery announcements / SPDP2 assertions	Find participants; maintain and clean up discovery records	Intended peers, cadence, expiry, and byte cost
Writer liveliness	Track an individual writer	Where it is needed; effect on filtering
RouterHealth	Periodic platform metadata and presence notifications	Intended monitors receive it; missed updates notify
Commands, reports, and status	Carry application work	Correct control/platform roles, teams, destinations, and sender-side filtering
Acknowledgments and repairs	Support reliable delivery	Expected return paths and traffic during loss / recovery

Validate with wire evidence, not just correct application output.

Keep discovery and system presence separate

1. **SPDP2:** maintain the discovery database and clean up expired participants.
2. **RouterHealth:** provide periodic metadata and callbacks for system presence.
3. **Application traffic:** keep writer-side filtering for the intended recipients.
4. **Wire validation:** account for every traffic type by purpose and system role.

> **The operational question is simple:**
>
> "Are we still hearing from the platform, or should we report it down?"

Connext 7.7 references: writer liveliness | SPDP2 | deadline callbacks